Evaluation of a real-time polymerase chain reaction (PCR) assay for detection of anisakis simplex parasite as a food-borne allergen source in seafo...

Anisakis simplex has been recognized as an important cause of disease in humans and as a food-borne allergen source. Actually, this food-borne parasite was recently identified as an emerging food safety risk. An A. simplex -specific primer-probe system based on a real-time polymerase chain reaction (PCR) detection assay has been successfully optimized and validated with seafood samples. In addition, a DNA extraction procedure has been optimized to detect the presence of the nematode in food samples. The assay is a very reliable, specific, and sensitive methodology to detect the presence of traces of this parasite in seafood products, including highly processed samples. As a result, 13 sequences of cytochrome c oxidase II gene were obtained and scrutinized to calculate intra- and interspecific variabilities of 0 and 35-67%, respectively. Finally, an efficiency of 2.07 +/- 0.14 of the assay was calculated, and a limit of detection of 40 ppm parasite in 25 g of sample was also optimized. Actually, the presence of this parasite in several seafood products has been demonstrated, enforcing the necessity of a design for a good manufacturing practice protocol for the processing industry to minimize the presence of this parasite as a food-borne allergen source in seafood products.